Banothu Sahasra

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.93
Narayana Junior College Kamalanagar	Class XII (TGBIE)	2024	94.40%
Dhanbad Public School Po K G Ashram	Class X (CBSE)	2022	84.00%

POSITIONS OF RESPONSIBILITY

Campus Tour Coordinator | PRAYAS, Rural Development Centre (RDC), IIT Hyderabad

- *Coordinated campus tours for school and college students visiting IIT Hyderabad*
- *Collaborated with volunteers to ensure a smooth and engaging visitor experience*

SKILLS

CAD & Simulation: Solid Edge · Abaqus

Programming & Analysis: MATLAB · Python · Excel

Engineering Tools: LTspice · LaTeX

RELEVANT COURSES

Mechanical Fundamentals: Engineering Mechanics · Mechanics of Solids · Thermodynamics · Fluid Mechanics

Design & Machine Dynamics: Kinematics & Dynamics of Machinery · Digital Fabrication

Manufacturing & Thermal Sciences: Manufacturing Science · Heat & Mass Transfer

PROJECTS

Interactive Slider-Crank Mechanism Simulator | *Python, NumPy, Matplotlib, Tkinter*

- Developed an interactive **slider-crank mechanism simulator** with user-defined geometric and kinematic parameters.
- Implemented **position, velocity, and acceleration analysis** for the primary slider-crank configuration.
- Built real-time mechanism animation and **position, velocity, and acceleration vector diagrams**.
- Developed a Tkinter-based GUI for parameter input, mechanism visualization, and numerical results.

Backhoe Loader - 3D CAD Modeling | *Solid Edge*

- Created a detailed **3D CAD model of a backhoe loader** using solid and feature-based modeling techniques.
- Modeled major machine components including the **chassis, cabin, wheels, front loader, and rear excavator**.
- Developed and refined **complex 3D geometry** in Solid Edge to create the complete machine model.

Nonlinear FEA of a Portal Frame | *Abaqus*

- Performed **geometrically and materially nonlinear FEA** of a portal frame using 1D beam elements and an elastic-perfectly plastic material model.
- Analyzed **load- and displacement-controlled responses**, identifying first yield, plastic hinge formation, and the sway collapse mechanism.
- Conducted **mesh sensitivity and convergence analysis**.